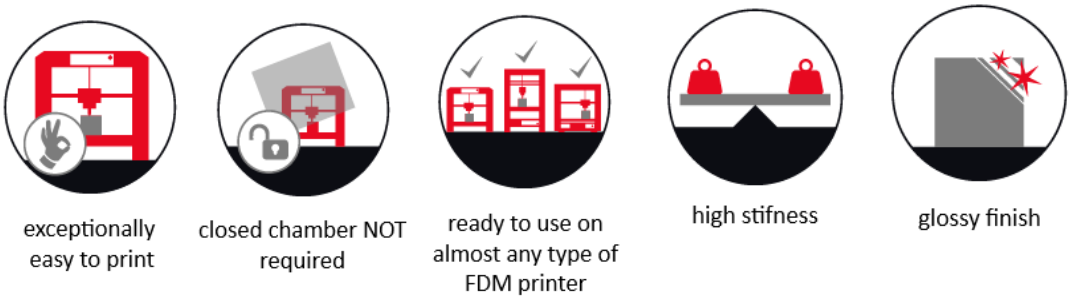
NOCTUO

PRODUCT CARD

COSMIC



1. GENERAL INFORMATION ABOUT THE PRODUCT

Cosmic is based on PLA, one of the most popular and common material in 3D printing. Thanks to addition of glittery elements, it is perfect for prints which require high esthetic level.

Main Cosmic features:

- very easy to print,
- ready to use in almost every FDM 3D Printer,
- high stiffness,
- glitter addition, which gives many possibilities to use prints in everyday life.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ASTM			
PHYSICAL				
Density	D792	-	g/cm^3	1.24
MECHANICAL				
Tensile strength	D882	-	MPa	60
Breaking strain	D882	-	MPa	53
Elongation at break	D882	-	%	6
Modulus of flexibility	D882	-	GPa	3,6
Bending Strength	D790	-	MPa	83
Modulus of elasticity	D790	-	GPa	3.8
Notch impact strength, Izod	D256	-	J/m	16
THERMAL				
Deflection temperature under load	E2092	0,45 MPa	°C	55

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

Cosmic does not require drying before using. Satisfying results could be achieved by using standard settings for PLA in available slicers and printers. It may require retraction adjustment.

Cooling during printing is recommended.

Due to additives, it is not recommended to use this filament in printers with a small nozzle (below 0,4mm).

Recommended parameters of printing:

Hotend temperature	200 - 230 °C
Bed temperature	50 - 60 °C
Print speed	< 300 mm/s

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
Cosmic needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

In proper using conditions, the product doesn't endanger health.

It's forbidden to set fire or exceed decomposition temperature!

Decomposition of PLA is typically over 250 °C and it should be strictly avoided.
Main ingredient of decomposition is lactide.

Detailed safety information available in SDS.